

When $E(\mathbb{Q})$ is not dense in $E(\mathbb{Q}_p)$

Six worked examples, one at each layer of the filtration $E(\mathbb{Q}_p) \supset E_0 \supset E_1 \supset E_2$ — including a curve where the reduction map $E(\mathbb{Q}) \rightarrow E(\mathbb{F}_p)$ is **surjective** and the image is still not dense

on a question of Rene Pannekoek; everything checked in `ec-padic-closure.gp`, output in `results/ec-padic-closure.txt`

The shortest answer. Let $E : y^2 + y = x^3 - x$ (conductor 37), of rank 1 with generator $P = (0, 0)$ and no torsion, and let $p = 23$. Then $E(\mathbb{F}_{23}) \cong \mathbb{Z}/22$ while $\overline{P}$ has order 11, so the closure of $E(\mathbb{Q})$ in $E(\mathbb{Q}_{23})$ has index 2. For a prime of **bad** reduction take $E : y^2 + y = x^3 + x^2$ (conductor 43) and $p = 43$: the reduction is non-split multiplicative, $E(\mathbb{Q}_{43})/E_1(\mathbb{Q}_{43})$ is cyclic of order $p + 1 = 44$, and $\overline{P}$ generates only the subgroup of order 22 — index 2 again.

1. The shape of the answer

For $p \geq 3$ the group $E(\mathbb{Q}_p)$ is a compact p -adic Lie group of dimension 1, and

$$E(\mathbb{Q}_p) \cong \mathbb{Z}_p \times T, \quad T \text{ finite abelian}.$$

Let $\Gamma \subseteq E(\mathbb{Q}_p)$ be the closure of the image of $E(\mathbb{Q})$. It is a closed subgroup, so:

Γ is open if and only if rank $E(\mathbb{Q}) \geq 1$. If the rank is 0 then $E(\mathbb{Q})$ is finite and Γ is finite, hence nowhere dense in the uncountable group $E(\mathbb{Q}_p)$ — at **every** p , for trivial reasons. If the rank is ≥ 1 then Γ is infinite; $\Gamma \cap (\mathbb{Z}_p \times 0)$ has finite index in Γ , hence is an infinite closed subgroup $p^k \mathbb{Z}_p$ of $\mathbb{Z}_p$, which is open. So Γ is open and $[E(\mathbb{Q}_p) : \Gamma]$ is **finite**.

Density is therefore not a delicate analytic condition but the vanishing of a computable finite index, and the interesting examples all have rank ≥ 1 . Every curve below has rank exactly 1 and trivial torsion, so $E(\mathbb{Q}) = \langle P \rangle$ and $\Gamma = \overline{\langle P \rangle}$.

1.1. The filtration

Write $\mathcal{E}$ for the Néron model over $\mathbb{Z}_p$ and

$$E(\mathbb{Q}_p) \supseteq E_0(\mathbb{Q}_p) \supseteq E_1(\mathbb{Q}_p) \supseteq E_2(\mathbb{Q}_p) \supseteq \dots,$$

where E_0 is the set of points with non-singular reduction and E_n the set of points reducing to O modulo p^n . The three layers are:

layer	what it is	order
E/E_0	the component group $\Phi(\mathbb{F}_p)$	c_p , the Tamagawa number
E_0/E_1	$\tilde{E}^{\text{ns}}(\mathbb{F}_p)$, the smooth part of the special fibre	$p + 1 - a_p$ (good); $p - a_p$ (bad)
E_1	the formal group, $\cong (\mathbb{Z}_p, +)$ for $p \geq 3$, with $E_n \leftrightarrow p^{n-1} \mathbb{Z}_p$	$E_1/E_2 \cong \mathbb{F}_p$

For bad p the middle row is $\mathbb{F}_p^\times$ (split multiplicative, $a_p = 1$), the norm-one torus of $\mathbb{F}_{p^2}/\mathbb{F}_p$ (non-split, $a_p = -1$), or $\mathbb{G}_a(\mathbb{F}_p)$ (additive, $a_p = 0$); in all cases at once its order is $p - a_p$ for bad p and $p + 1 - a_p$ for good p .

The one computational fact needed: for a rational point Q with $x(Q) = a/c^2$ in lowest terms, the formal parameter is $z = -x/y$ and

$$v_p(z(Q)) = v_p(c), \quad \text{so} \quad Q \in E_n(\mathbb{Q}_p) \iff p^n \mid c.$$

1.2. The index, and the criterion

Proposition (rank 1, trivial torsion, $p \geq 3$). Put $N_p = |E(\mathbb{Q}_p)/E_1(\mathbb{Q}_p)|$, let m be the order of $\overline{P}$ in that finite group — equivalently the least $m \geq 1$ with $p \mid c(mP)$ — and let $v = v_p(c(mP))$. Then

$$[E(\mathbb{Q}_p) : \Gamma] = (N_p/m) \cdot p^{v-1}.$$

Proof. $\Gamma \cap E_1 = \overline{\langle mP \rangle} = \mathbb{Z}_p \cdot (mP) = E_v(\mathbb{Q}_p)$, using $E_1 \cong \mathbb{Z}_p$ and $E_n \leftrightarrow p^{n-1}\mathbb{Z}_p$; and $\Gamma/(\Gamma \cap E_1) = \langle \overline{P} \rangle$ has index N_p/m in $E(\mathbb{Q}_p)/E_1$. Multiply the two indices, using $[E_1 : E_v] = p^{v-1}$. ■

The criterion in its clean form. For $p \geq 3$ and any rank,

$$\Gamma = E(\mathbb{Q}_p) \iff E(\mathbb{Q}) \rightarrow E(\mathbb{Q}_p)/E_2(\mathbb{Q}_p) \text{ is surjective,}$$

a finite check on a group of order $p \cdot N_p$.

Proof. ($\Leftarrow$) Suppose the map is onto. If $\Gamma \cap E_1 \subseteq E_2$ then $\Gamma \cap E_2 = \Gamma \cap E_1$, so the image of Γ in E/E_2 has order $|\Gamma/(\Gamma \cap E_1)| \leq N_p < pN_p$ — not onto. Hence $\Gamma \cap E_1 = E_1$, i.e. $\Gamma \supseteq E_1$, and then surjectivity onto E/E_2 forces $\Gamma/E_1 = E/E_1$. ($\Rightarrow$) is clear. ■

A caveat worth stating. For p of good reduction, surjectivity of the reduction map $E(\mathbb{Q}) \rightarrow E(\mathbb{F}_p)$ is **necessary** for density but it is **not sufficient**: it only controls the top two layers, and says nothing about E_1/E_2 . Section 2.3 gives a curve of conductor 89 and two primes, $p = 11$ and $p = 13$, where $E(\mathbb{Q}) \rightarrow E(\mathbb{F}_p)$ is onto and the closure still has index p . The correct statement is the boxed criterion above: one needs surjectivity modulo p^2 , not modulo p .

(For $p = 2$ the formal group need not be $\cong \mathbb{Z}_2$ — $E_1(\mathbb{Q}_2)$ can have 2-torsion, only $E_2(\mathbb{Q}_2) \cong \mathbb{Z}_2$ is automatic — so everything below takes p odd.)

2. Good reduction

2.1. The reduction map is not surjective: conductor 37, $p = 23$

$$E : y^2 + y = x^3 - x, \quad \Delta = 37, \quad \text{rank} = 1, \quad E(\mathbb{Q})_{\text{tors}} = 0, \quad P = (0, 0).$$

PARI's `ellrank` returns this generator and `ellsaturation` confirms it is saturated. At $p = 23$ the reduction is good, $a_{23} = 2$, and

$$E(\mathbb{F}_{23}) \cong \mathbb{Z}/22, \quad \text{ord}(\overline{P}) = 11, \quad [E(\mathbb{F}_{23}) : \langle \overline{P} \rangle] = 2.$$

Then $11P = (116/529, -8555/12167)$ has $c = 23$, so $v = 1$ and $11P \in E_1 \setminus E_2$: the formal layer is full. The index is $2 \cdot 23^0 = 2$.

The closure of $E(\mathbb{Q})$ in $E(\mathbb{Q}_{23})$ is an open subgroup of index 2. Concretely, $E(\mathbb{Q}_{23}) \cong \mathbb{Z}_{23} \times \mathbb{Z}/22$ and $\Gamma \cong \mathbb{Z}_{23} \times 2\mathbb{Z}/22$: no rational point is 23-adically close to a point whose reduction is a generator of $E(\mathbb{F}_{23})$.

Here is the index at every good $p < 100$ for this curve. It is 1 exactly when $\overline{P}$ generates $E(\mathbb{F}_p)$, which fails often:

p	3	5	7	11	13	17	19	23	29	31	41
index	1	1	1	1	1	1	1	2	2	2	1

p	43	47	53	59	61	67	71	73	79	83	89
index	3	1	1	4	2	2	1	1	1	3	1

2.2. A structural reason: $E(\mathbb{F}_p)$ non-cyclic, conductor 37, $p = 67$

If $E(\mathbb{Q})$ has rank 1 and no torsion then its image in $E(\mathbb{F}_p)$ is **cyclic**, so surjectivity is impossible as soon as $E(\mathbb{F}_p)$ is not. At $p = 67$,

$$E(\mathbb{F}_{67}) \cong \mathbb{Z}/30 \times \mathbb{Z}/2,$$

so the image can never be everything, whatever the generator is. The index is 2. For this curve the good $p < 400$ with $E(\mathbb{F}_p)$ non-cyclic are

$$p = 67, 107, 137, 139, 151, 233, 269, 293, 317, 349, 367,$$

and over all 667 good $p < 5000$, $E(\mathbb{F}_p)$ is non-cyclic for 113 of them (16.9%) — so this reason alone already forces non-density at a positive proportion of primes.

2.3. Surjective reduction, still not dense: conductor 89, $p = 11$ and 13

This is the example that separates the two criteria.

$$E : y^2 + xy + y = x^3 + x^2 - x, \quad \Delta = -89, \quad \text{rank} = 1, \quad E(\mathbb{Q})_{\text{tors}} = 0, \quad P = (0, 0).$$

At $p = 11$: $a_{11} = -2$, $E(\mathbb{F}_{11}) \cong \mathbb{Z}/14$, and $\bar{P}$ has order 14. **The reduction map $E(\mathbb{Q}) \rightarrow E(\mathbb{F}_{11})$ is surjective.** Nevertheless

$$14P = (2541600351/1274133025, 90518784046991/45480178327375),$$

whose x -denominator is c^2 with

$$c = 35695 = 5 \cdot 11^2 \cdot 59, \quad \text{so} \quad v_{11}(c) = 2 \quad \text{and} \quad 14P \in E_2(\mathbb{Q}_{11}).$$

Hence $\Gamma \cap E_1 = E_2(\mathbb{Q}_{11})$ and $[E(\mathbb{Q}_{11}) : \Gamma] = 1 \cdot 11^1 = 11$.

The same curve does it again at $p = 13$: $E(\mathbb{F}_{13}) \cong \mathbb{Z}/12$ is generated by $\bar{P}$, and $12P = (-8703421/5597956, -3467599409/13244763896)$ has $c = 2366 = 2 \cdot 7 \cdot 13^2$, so $12P \in E_2(\mathbb{Q}_{13})$ and the index is 13.

What is going on. $E(\mathbb{Q}) \rightarrow E(\mathbb{F}_p)$ being onto says the closure surjects onto the top two layers. It says nothing about $E_1/E_2 \cong \mathbb{F}_p$. Here the whole group $E(\mathbb{Q}) \cap E_1(\mathbb{Q}_p) = \langle 14P \rangle$ already lies in E_2 , so its closure is $E_2(\mathbb{Q}_p) = p\mathbb{Z}_p$ and misses an index- p amount of the formal group. The rational points surject onto $E(\mathbb{F}_{11})$ but not onto $E(\mathbb{Q}_{11})/E_2(\mathbb{Q}_{11})$, which is the group of order $11 \cdot 14 = 154$ that actually decides density.

Heuristically $v_p(c(mP)) \geq 2$ has “probability” $1/p$, so this is rare but not at all exceptional — and it happens twice on this one curve, at the two smallest primes where it could.

3. Bad reduction

3.1. The torus layer: conductor 43, $p = 43$

$$E : y^2 + y = x^3 + x^2, \quad \Delta = -43, \quad \text{rank} = 1, \quad E(\mathbb{Q})_{\text{tors}} = 0, \quad P = (0, 0).$$

At $p = 43$ the reduction is multiplicative of type I_1 , so $c_{43} = 1$ and the component group is trivial; $a_{43} = -1$, so the reduction is **non-split** and

$$E_0(\mathbb{Q}_{43})/E_1(\mathbb{Q}_{43}) \cong \ker(N : \mathbb{F}_{43^2}^\times \rightarrow \mathbb{F}_{43}^\times) \cong \mathbb{Z}/44.$$

The singular point of the reduction is $(28, 21)$ and $P = (0, 0)$ does not meet it, so $P \in E_0(\mathbb{Q}_{43})$. Its order in $E(\mathbb{Q}_{43})/E_1$ is 22, exactly half of 44; and $22P$ has $c = 3033521 = 19 \cdot 43 \cdot 47 \cdot 79$, so $v = 1$ and the formal layer is full. The index is 2.

This is the bad-reduction twin of Section 2.1: the finite layer is a **torus** rather than $E(\mathbb{F}_p)$, but the mechanism — a generator landing in an index-2 subgroup of a cyclic group of even order — is identical.

3.2. The formal layer: conductor 77, $p = 11$

$$E : y^2 + y = x^3 + 2x, \quad \Delta = -539 = -7^2 \cdot 11, \quad \text{rank} = 1, \quad E(\mathbb{Q})_{\text{tors}} = 0, \quad P = (2, -4).$$

At $p = 11$: type I_1 , $c_{11} = 1$, $a_{11} = -1$, so $E(\mathbb{Q}_{11})/E_1(\mathbb{Q}_{11})$ is cyclic of order 12 — and $\bar{P}$ **generates all of it**. But

$$12P = (-89535/937024, -235033185/907039232), \quad c = 968 = 2^3 \cdot 11^2,$$

so $12P \in E_2(\mathbb{Q}_{11})$ and the index is 11. This is the bad-reduction twin of Section 2.3.

3.3. The component layer: conductor 11289, $p = 3$

The one mechanism with no good-reduction analogue: missing a whole component of the Néron model.

$$E : y^2 + xy = x^3 + x^2 - 9, \quad \Delta = -33867 = -3^2 \cdot 53 \cdot 71, \quad \text{rank} = 1, \quad E(\mathbb{Q})_{\text{tors}} = 0, \quad P = (2, 1).$$

At $p = 3$ the reduction is of type I_2 , non-split, with

$$c_3 = 2, \quad \Phi(\mathbb{F}_3) \cong \mathbb{Z}/2, \quad E_0/E_1 \cong \mathbb{Z}/4, \quad |E(\mathbb{Q}_3)/E_1(\mathbb{Q}_3)| = 2 \cdot 4 = 8.$$

The singular point of the reduction mod 3 is $(0, 0)$, and $P = (2, 1)$ reduces to $(2, 1) \neq (0, 0)$. So $P \in E_0(\mathbb{Q}_3)$, and since $E_0(\mathbb{Q}_3)$ is open **and closed**,

$$\Gamma \subseteq E_0(\mathbb{Q}_3).$$

The image of $E(\mathbb{Q})$ in the component group $\Phi(\mathbb{F}_3) \cong \mathbb{Z}/2$ is **trivial**: no rational point at all lies on the non-identity component of the special fibre. The order of $\bar{P}$ in $E(\mathbb{Q}_3)/E_1$ is $4 = |E_0/E_1|$, confirming that the whole loss at that stage is the component group; and $4P$ has $c = 24552 = 2^3 \cdot 3^2 \cdot 11 \cdot 31$, so $v = 2$ and the formal group contributes another factor 3. Total index $2 \cdot 3 = 6$.

3.4. How common is the component layer, really

An earlier draft of these notes called this mechanism rare. That was wrong, and the way it was wrong is worth recording: the claim rested on a listing that had been truncated to its first 28 rows, and on a sweep that excluded $p = 2$. A proper sweep of 156450 curves in reduced minimal form ($a_1, a_3 \in \{0, 1\}$, $a_2 \in \{-1, 0, 1\}$, $|a_4| \leq 40$, $|a_6| \leq 80$) turns up 5249 confirmed pairs (E, p) with certified rank ≥ 1 and **proper** image in $\Phi(\mathbb{F}_p)$, against 60022 pairs with $c_p \geq 2$ in the same box. (That denominator counts rank-0 curves too, which are excluded from the numerator, so the rate among rank- ≥ 1 curves is higher than the 8.7% this ratio suggests.) The mechanism is ordinary, not exceptional.

Two things had hidden it. First, 3060 of the 5249 sit at $p = 2$, which the sweeps behind the earlier draft skipped — and $p = 2$ is perfectly legitimate here: the component argument needs only that $E_0(\mathbb{Q}_p)$ is open **and closed**, so unlike the formal-group layer of Section 1 it has no $p = 2$ caveat at all. Second, the surviving examples were sorted by conductor and then read off the top of a truncated list.

The smallest conductors found, over all minimal models of conductor ≤ 600 in a box with $|a_4| \leq 150$, $|a_6| \leq 350$:

cond.	curve	p	type	c_p	image	rank, tors.
438	$y^2 + xy = x^3 + x^2 - 65x - 231$	2	I_2	2	1	1, $\mathbb{Z}/2$
446	$y^2 + xy + y = x^3 + x^2 - 39x - 35$	2	I_{14}	14	7	1, 0

The second is the more striking of the two. It has **split** multiplicative reduction of type I_{14} at $p = 2$, so $\Phi(\mathbb{F}_2) \cong \mathbb{Z}/14$ — fourteen components — and rank 1 with trivial torsion and saturated generator $P = (-5, 10)$. The singular point of the reduction mod 2 is $(1, 0)$, and kP avoids it exactly for $k = 7$ and $k = 14$:

$$7P = (-25/4, 47/8) \in E_1(\mathbb{Q}_2), \quad 14P = (1139761/10816, -1281713061/1124864) \in E_1(\mathbb{Q}_2),$$

while kP reduces to $(1, 0)$ for every other $k \leq 15$. So the image of P in $\mathbb{Z}/14$ has order 7: **$E(\mathbb{Q})$ lands in the even part $2\mathbb{Z}/14\mathbb{Z}$ and misses all seven odd components of the Néron model outright.** The closure has index divisible by 2.

4. How often, and the degenerate case

For the conductor-37 curve, over the 667 primes of good reduction below 5000:

$E(\mathbb{Q}) \rightarrow E(\mathbb{F}_p)$ not surjective	357 primes, 53.5%
of which forced, $E(\mathbb{F}_p)$ non-cyclic	113 primes, 16.9%
$E(\mathbb{Q}) \rightarrow E(\mathbb{F}_p)$ surjective	310 primes, 46.5%

So non-density is the **typical** behaviour, not a curiosity: it happens for more than half the primes. The density of p for which a fixed point of infinite order generates $E(\mathbb{F}_p)$ is the elliptic analogue of Artin’s primitive-root problem, studied by Lang and Trotter; the 46.5% above is an observed value of that density, not a theorem.

And the degenerate case, worth naming so it is not mistaken for the phenomenon: if $\text{rank } E(\mathbb{Q}) = 0$ — for instance $y^2 + y = x^3 - x^2 - 10x - 20$ of conductor 11, with $E(\mathbb{Q}) \cong \mathbb{Z}/5$ — then $E(\mathbb{Q})$ is finite and $E(\mathbb{Q}_p)$ contains $E_1(\mathbb{Q}_p) \cong \mathbb{Z}_p$, so the image is nowhere dense at every single prime. Every example above has rank exactly 1 precisely to avoid this.

5. Summary

curve (conductor)	p	reduction	layer that fails	$E(\mathbb{Q}) \rightarrow E(\mathbb{F}_p)$ onto?	index
$y^2 + y = x^3 - x$ (37)	23	good	E_0/E_1	no	2
$y^2 + y = x^3 - x$ (37)	67	good, $E(\mathbb{F}_p)$ non-cyclic	E_0/E_1	impossible	2
$y^2 + xy + y = x^3 + x^2 - x$ (89)	11	good	E_1/E_2	yes	11
$y^2 + xy + y = x^3 + x^2 - x$ (89)	13	good	E_1/E_2	yes	13
$y^2 + y = x^3 + x^2$ (43)	43	non-split mult. I_1	E_0/E_1	—	2
$y^2 + y = x^3 + 2x$ (77)	11	non-split mult. I_1	E_1/E_2	—	11
$y^2 + xy = x^3 + x^2 - 9$ (11289)	3	non-split mult. I_2	E/E_0 and E_1/E_2	—	6

The closure of $E(\mathbb{Q})$ in $E(\mathbb{Q}_p)$ is a closed subgroup, open of finite index as soon as the rank is positive, so “dense” is a finite computation. It fails whenever the image misses any layer of $E(\mathbb{Q}_p) \supset E_0 \supset E_1 \supset E_2$: a component of the Néron model, part of the reduction $\tilde{E}^{\text{ns}}(\mathbb{F}_p)$, or a step of the formal group. The first is rare, the second is the usual reason and happens for over half of all p , and the third is the one that the criterion “ $E(\mathbb{Q}) \rightarrow E(\mathbb{F}_p)$ is surjective” does not see — which is why the correct statement is surjectivity onto $E(\mathbb{Q}_p)/E_2(\mathbb{Q}_p)$, i.e. modulo p^2 .

Companion file: `ec-padic-closure.gp`, run as

```
gp -q -s 4000000000 ec-padic-closure.gp < /dev/null > results/ec-padic-closure.txt
```

It recomputes every number above: the ranks and saturated generators via `ellrank` and `ellsaturation`, the group structures via `ellgroup`, the local data via `elllocalred` and `ellap`, the singular points of the bad reductions by brute force, the exact multiples mP over $\mathbb{Q}$, and the index $(N_p/m)p^{v-1}$ at each place — together with the two sweeps of Section 2.1 and Section 4.